

Classes and Objects Practice Questions

Question 1

Your school wants a simple program to store and display student information. Create a Student class that can hold a student's details and greet them.

YOUR TASKS

- Create a class called Student
- Add three attributes: name (String), age (int), grade (String)
- Write a method displayInfo() that prints all three attributes
- Write a method greet() that prints a welcome message using the student's name
- In main, create two Student objects with different values and call both methods on each

SAMPLE OUTPUT

```
Name: Alice
```

```
Age: 14
```

```
Grade: 8A
```

```
Welcome to school, Alice!
```

```
Name: Ben
```

```
Age: 15
```

```
Grade: 9B
```

```
Welcome to school, Ben!
```

Question 2

A local pet clinic needs to track animals brought in for checkups. Create a Pet class that stores each animal's details and can check if it needs a vaccine based on its age.

YOUR TASKS

- Create a class called Pet
- Add three attributes: name (String), type (String), age (int)
- Write a method displayInfo() that prints all attributes
- Write a method needsVaccine() — if age is less than 2, print that the pet needs a vaccine; otherwise print that it is up to date
- In main, create three Pet objects and call both methods on each

SAMPLE OUTPUT

```
Name: Milo   |   Type: Dog   |   Age: 1
Milo needs a vaccine!
```

```
Name: Luna   |   Type: Cat   |   Age: 3
Luna is up to date.
```

```
Name: Kiwi   |   Type: Rabbit  |   Age: 1
Kiwi needs a vaccine!
```

Question 3

A small online shop wants to keep track of its products. Create a Product class that stores product details and can apply a discount or check if the item is in stock.

YOUR TASKS

- Create a class called Product
- Add four attributes: name (String), price (double), quantity (int), category (String)
- Write a method displayInfo() that prints all attributes
- Write a method applyDiscount(double percent) that reduces the price by that percentage and prints the new price
- Write a method isInStock() — if quantity is greater than 0, print in stock; otherwise print out of stock
- In main, create two Product objects, display their info, apply a discount to one, and check stock for both

Sample output

Name: Headphones

Price: \$49.99

Quantity: 10

Category: Electronics

Headphones is in stock.

Discount applied! New price: \$39.99

Name: Notebook

Price: \$5.00

Quantity: 0

Category: Stationery

Notebook is out of stock.